Python: exercises on iteration

This notebook was generated in pseudocode mode.

Exercise: Print Numbers from 1 to N

Exercise Description

Write a function that prints all numbers from 1 to N using a for loop and range.

Your solution

```
def print_numbers(n):
    # Step 1: Set up a for loop to iterate from 1 to n (inclusive)
    # Step 2: For each number in the range, print it
```

Test your solution

```
# Test case 1: Test with small number
print_numbers(5)
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Sum of First N Natural Numbers

Exercise Description

Write a function to calculate the sum of the first N natural numbers (1 + 2 + 3 + ... + N) using a while loop.

Your solution

```
def sum_natural_numbers(n):
    # Step 1: Initialize sum variable to 0 and counter to 1
    # Step 2: Set up while loop that continues while counter <= n
    # Step 3: Add current counter value to sum
    # Step 4: Increment counter by 1
    # Step 5: Return the final sum
```

Test your solution

```python
# Test case 2: Test with another number
result = sum_natural_numbers(3)
expected = 1 + 2 + 3  # 6
assert result == expected, f"Expected {expected}, got {result}"
print(f"✓ sum_natural_numbers(3) = {result}")
```

Test your solution

```python
# Test case: Example from solution
result = sum_natural_numbers(5)
print(f"Sum of first 5 natural numbers: {result}")  # 15
```

Debug your solution

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```python
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Print Even Numbers from 2 to 20

Exercise Description

Write a function that prints all even numbers between 2 and 20 using a for loop and range.

Your solution

```
def print_even_numbers():
    # Step 1: Set up a for loop with start=2, stop=21, step=2
    # Step 2: For each even number in the range, print it
```

Test your solution

```
# Test case 2: Verify range and step
# Expected: prints 2, 4, 6, 8, 10, 12, 14, 16, 18, 20 (each on new line)
print_even_numbers()
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Count Down from N to 1 (If Even)

Exercise Description

Write a function to print numbers from N to 1 using a while loop, but only if N is even.

Your solution

```
def countdown_if_even(n):
    # Step 1: Check if n is even using modulo operator
    # Step 2: If n is even, set up while loop that continues while n > 0
    # Step 3: Print current n value
    # Step 4: Decrement n by 1
    # Step 5: If n is odd, do nothing
```

Test your solution

```
# Test case 2: Test with odd number (should print nothing)
countdown_if_even(3)
```

Test your solution

```
# Test case 2 (assert)
countdown_if_even(6)  # Prints 6, 5, 4, 3, 2, 1
```

Debug your solution

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Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Multiplication Table of 3

Exercise Description

Write a function that prints the multiplication table of 3 (from 3x1 to 3x10) using a for loop.

Your solution

```
def multiplication_table_3():
    # Step 1: Set up a for loop to iterate from 1 to 10
    # Step 2: For each multiplier, calculate 3 * multiplier
    # Step 3: Print the multiplication equation and result
```

Test your solution

```
# Test case 2: Verify format
# Expected: prints 3 x 1 = 3, 3 x 2 = 6, ..., 3 x 10 = 30
multiplication_table_3()
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Sum of First N Even Numbers

Exercise Description

Write a function to calculate the sum of the first N even numbers (2 + 4 + 6 + ...) using a while loop.

Your solution

```
def sum_first_n_even(n):
    # Step 1: Initialize sum to 0, current even number to 2, and count to 0
    # Step 2: Set up while loop that continues while count < n
    # Step 3: Add current even number to sum
    # Step 4: Move to next even number (add 2)
    # Step 5: Increment count
    # Step 6: Return the final sum
```

Test your solution

```python
# Test case 2: Test with another number
result = sum_first_n_even(5)
expected = 2 + 4 + 6 + 8 + 10  # 30
assert result == expected, f"Expected {expected}, got {result}"
print(f"✓ sum_first_n_even(5) = {result}")
```

Debug your solution

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```python
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Numbers Divisible by 5 (1 to 50)

Exercise Description

Write a function that prints all numbers divisible by 5 between 1 and 50 using a for loop and range.

Your solution

```python
def print_divisible_by_5():
    # Step 1: Set up a for loop with start=5, stop=51, step=5
```

```
    # Step 2: For each number divisible by 5, print it
```

Test your solution

```
# Test case 2: Verify range
# Expected: prints 5, 10, 15, 20, 25, 30, 35, 40, 45, 50
print_divisible_by_5()
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Factorial of 5

Exercise Description

Write a function to find the factorial of 5 using a while loop.

Your solution

```
def factorial_5():
    # Step 1: Initialize factorial to 1 and counter to 5
    # Step 2: Set up while loop that continues while counter > 0
    # Step 3: Multiply factorial by current counter value
    # Step 4: Decrement counter by 1
    # Step 5: Return the final factorial
```

Test your solution

```
# Test case 2: Verify result
result = factorial_5()
assert result == 120, f"Expected 120, got {result}"
print(f"✓ Factorial of 5 is {result}")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Sum of Odd Numbers (1 to 20)

Exercise Description

Write a function to calculate the sum of all odd numbers between 1 and 20 using a for loop.

Your solution

```
def sum_odd_numbers():
    # Step 1: Initialize sum to 0
    # Step 2: Set up for loop with start=1, stop=21, step=2 (odd numbers)
    # Step 3: Add each odd number to sum
    # Step 4: Return the final sum
```

Test your solution

```
# Test case 2: Verify result
result = sum_odd_numbers()
assert result == 100, f"Expected 100, got {result}"
print(f"✓ Sum of odd numbers 1-20 is {result}")
```

Test your solution

```
# Test case: Example from solution
result = sum_odd_numbers()
print(f"Sum of odd numbers 1-20: {result}")  # 100
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Numbers Divisible by K (1 to N)

Exercise Description

Write a function to print numbers between 1 and N that are divisible by K using a for loop and range.

Your solution

```
def print_divisible_by_k(n, k):
    # Step 1: Set up for loop with start=k, stop=n+1, step=k
    # Step 2: For each number divisible by k, print it
```

Test your solution

```
# Test case 2: Test with n=20, k=4
# Expected: prints 4, 8, 12, 16, 20
print_divisible_by_k(20, 4)
```

Test your solution

```
# Test case: Example from solution
print_divisible_by_k(20, 3)  # Prints 3, 6, 9, 12, 15, 18
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Count Digits in a Number

Exercise Description

Write a function that counts how many digits are in a positive integer.

Your solution

```
def count_digits(num):
    # Step 1: Initialize a counter to 0
```

```
    # Step 2: While num > 0, do integer division by 10 and increment counter
    # Step 3: Return the counter
```

Test your solution

```
assert count_digits(123456) == 6
assert count_digits(7) == 1
assert count_digits(10) == 2
print("✓ count_digits tests passed")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Cubes from 1 to 5

Exercise Description

Write a function that returns a list with the cubes of numbers from 1 to 5 using a for loop.

Your solution

```
def cubes_1_to_5():
    # Step 1: Initialize an empty list
    # Step 2: For i from 1 to 5 inclusive, append i**3 to the list
    # Step 3: Return the list
```

Test your solution

```
assert cubes_1_to_5() == [1, 8, 27, 64, 125]
print("✓ cubes_1_to_5 tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Product of Numbers from 1 to N

Exercise Description

Write a function that returns the product of numbers from 1 to N using a for loop.

Your solution

```
def product_1_to_n(n):
    # Step 1: Initialize product to 1
    # Step 2: For i from 1 to n inclusive, multiply product by i
    # Step 3: Return product
```

Test your solution

```
assert product_1_to_n(1) == 1
assert product_1_to_n(3) == 6
assert product_1_to_n(5) == 120
print("✓ product_1_to_n tests passed")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Odd Numbers from 1 to 15

Exercise Description

Write a function that returns all odd numbers from 1 to 15 using a while loop.

Your solution

```
def odd_1_to_15():
    # Step 1: Initialize i to 1 and an empty list
    # Step 2: While i <= 15, append i then increment by 2
    # Step 3: Return the list
```

Test your solution

```
assert odd_1_to_15() == [1, 3, 5, 7, 9, 11, 13, 15]
print("✓ odd_1_to_15 tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Print Numbers from 5 to 1

Exercise Description

Write a function that returns numbers from 5 to 1 using a for loop with range.

Your solution

```
def countdown_5_to_1():
    # Step 1: Initialize an empty list
    # Step 2: Use range from 5 down to 1 inclusive and append to list
    # Step 3: Return the list
```

Test your solution

```
assert countdown_5_to_1() == [5, 4, 3, 2, 1]
print("✓ countdown_5_to_1 tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Sum of the First 7 Odd Numbers

Exercise Description

Write a function to find the sum of the first 7 odd numbers using a while loop.

Your solution

```
def sum_first_7_odds():
    # Step 1: Initialize i=1, count=0, total=0
    # Step 2: While count < 7, add i to total, increment i by 2 and count by 1
    # Step 3: Return total
```

Test your solution

```
assert sum_first_7_odds() == 49
print("✓ sum_first_7_odds tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the

Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Repeat Until ”stop”

Exercise Description

Write a function that, given a sequence of strings, counts how many entries occur until the string "stop" appears.

Your solution

```
def until_stop_count(inputs):
    # Step 1: Initialize a counter to 0
    # Step 2: For each string in inputs:
    #    If the string is "stop", break the loop
    #    Otherwise increment the counter
    # Step 3: Return the counter
```

Test your solution

```
assert until_stop_count(["a", "b", "stop", "x"]) == 2
assert until_stop_count(["stop"]) == 0
```

```
assert until_stop_count(["hello", "world"]) == 2
print("✓ until_stop_count tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Repeat String as Many Times as Its Length

Exercise Description

Write a function that takes a string and returns a list with the string repeated as many times as its length.

Your solution

```
def repeat_string_len_times(s):
    # Step 1: Initialize an empty list
    # Step 2: For i in range(len(s)), append s
    # Step 3: Return the list
```

Test your solution

```
assert repeat_string_len_times("abc") == ["abc", "abc", "abc"]
assert repeat_string_len_times("") == []
print("✓ repeat_string_len_times tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Ask for a Password

Exercise Description

Write a function that, given a sequence of password attempts, returns the number of attempts needed to enter the correct password ("password123"). Return -1 if never entered.

Your solution